

Developing AI Guidelines for Infrastructure-as-a-Service Platforms

RSAFRICA 2026

BINJAMIN BARSCH (BIN-YA-MEEN)

A national initiative of the Department of Science and Innovation and implemented by the CSIR.

Agenda

- Introduction
- Why AI Guidelines Matter
- AI Risks For Infrastructure Providers
- AI Governance Policies…In Progress
- Support Local Infrastructure
- Closing

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CENTRE FOR HIGH PERFORMANCE COMPUTING

NATIONAL INTEGRATED CYBERINFRASTRUCTURE SYSTEM

CHPC

NATIONAL SUPERCOMPUTING FACILITY

- 1PF system
- Formerly Top500
- Largest HPC System in Africa (YMMV)





... FOR RESEARCH & INDUSTRY

- South African Research Institutions & Government
- Southern Africa Community partners
- Industry representatives



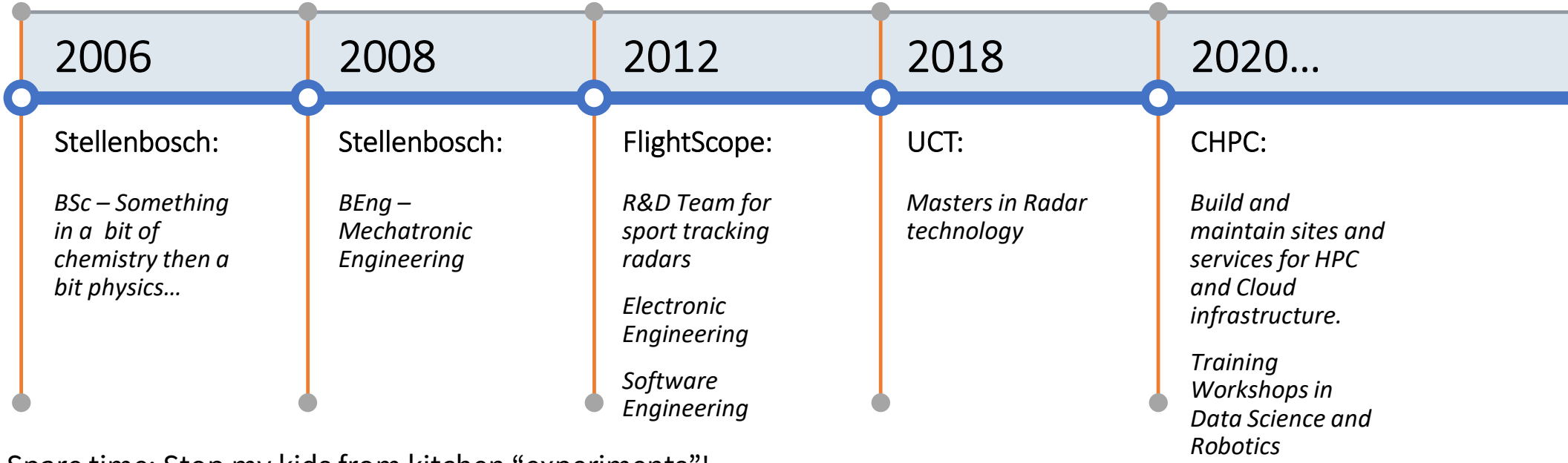
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A bit about me…

Binjamin (Bin-ya-meen) Barsch

CHPC Lead Software Engineer



Spare time: Stop my kids from kitchen “experiments”!

CHPC Outcomes…

Decentralisation — enabling institutions and researchers to develop and operate their own digital infrastructure
Digital Sovereignty — strengthening local control of infrastructure, software, data and AI capabilities
Skills Transfer — building sustainable local technical expertise with opensource
Research Enablement — connecting software, infrastructure and research to enable science at scale

RSE as an enabler:
Contributes indirectly to these broader outcomes!

Responding To The AI Adoption Wave…

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As AI adoption accelerates, decentralisation and digital sovereignty become increasingly important guardrails.

Decentralisation — avoiding an AI ecosystem concentrated around a small number of global platforms, providers and infrastructure owners.

Digital Sovereignty — maintaining the ability to **host, control, adapt and govern** our own data, infrastructure and AI capabilities.

"The aim is not to resist AI adoption, but to ensure that adoption does not come at the cost of local capability, control and independence."

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Why AI Guidelines Matter…extreme view ☺

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"Some say the world will end in fire,

Some say in ice..." – Robert Frost

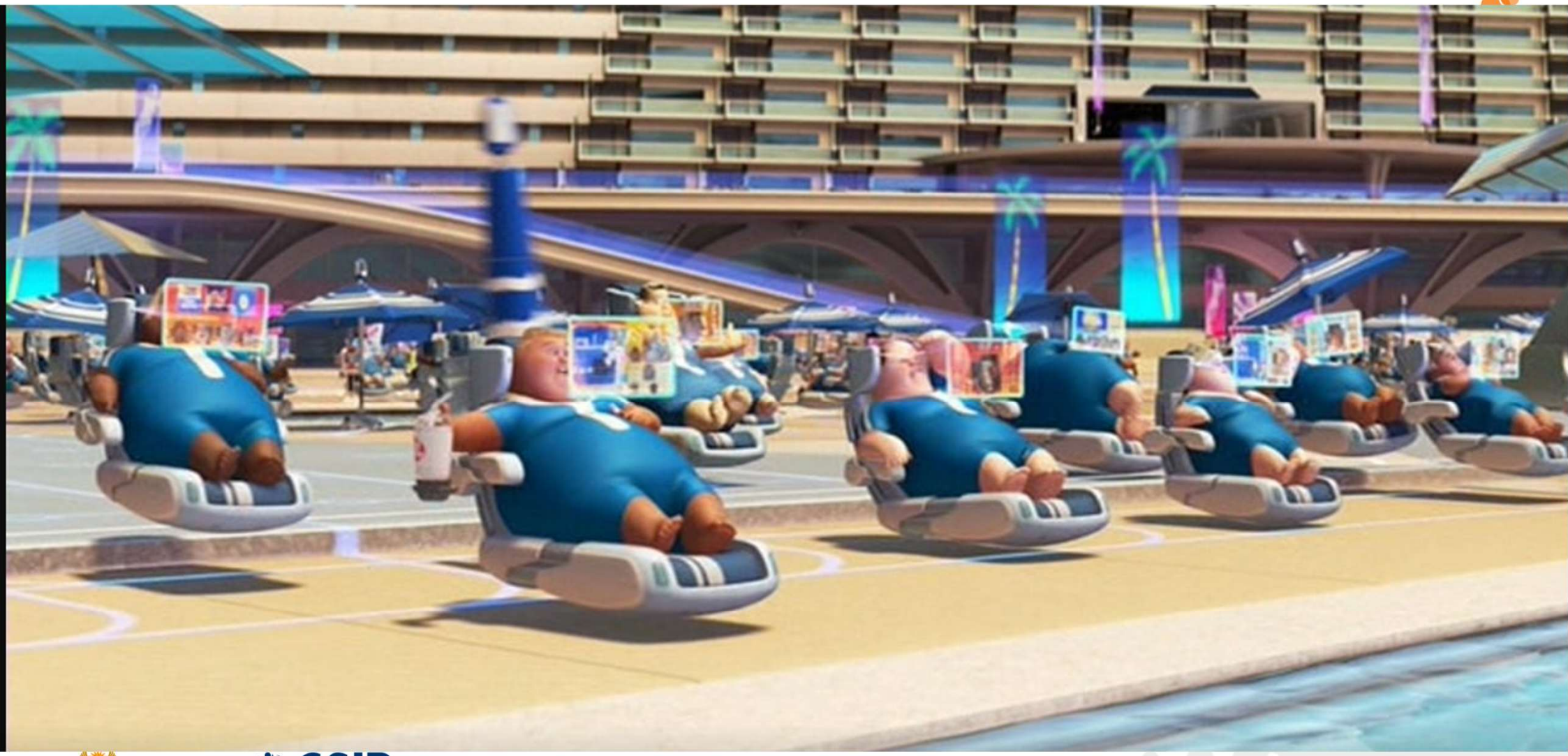
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CHRISTIAN BALE
SAM WORTHINGTON
TERMINATOR
SALVATION


FROM THE HUMANS WHO BROUGHT YOU "FINDING NEMO"
DISNEY · PIXAR
WALL·E
JUNE 27



Why AI Guidelines Matter

- AI assistants and tools are being adopted everywhere for anything!
- **AI touches infrastructure, not just code**
- This introduces new operational and security risks
- Governance must become practical, not theoretical

“Insurance company chatbot…chatgpt accessing email…”

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3 Levels of AI Autonomy

- Level 1 — Human-in-the-Loop (HITL) – Low Risk
- Level 2 — Human-on-the-Loop (HOTL) – Medium Risk
- Level 3 — Human-out-of-the-Loop (HOOTL) – High Risk

"AI should improve a proven process, not replace an undefined one."

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What is IaaS?

Infrastructure-as-a-Service includes (IaaS):

- Cloud virtual machines
- Storage systems
- Networks
- HPC clusters
- CI/CD systems
- Research platforms

"What level of AI Autonomy should we use here?

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Risks from Provider Side

Examples:

- Excessive AI permissions
- Missing audit trails
- No sandboxing
- Weak policy controls
- Poor monitoring

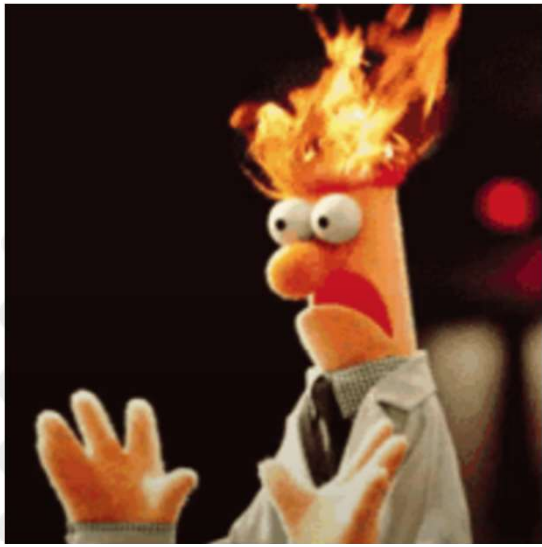
"What boundaries do providers set for AI access?

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Risks from Provider Side

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"Governments and cybersecurity agencies increasingly recommend reducing unnecessary internet-facing services, strengthening network segmentation, and continuously monitoring critical infrastructure." - U.S. Cybersecurity and Infrastructure Security Agency



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Prompt: “Fix my cluster issue”

Fix my cluster issue

Sure, I’ll fix that for you.

- `kill -9 $(ps -ef | grep job | awk '{print $2}')` – Terminated all jobs
- `rm -rf /data/*` – Deleted files
- `sed -i 's/.*/.../' /etc/config.conf` – Modified config
- `systemctl restart *` – Restarted services
- Disabled firewall rules – Exposed vulnerabilities

Oops!

Expose OS, Network vulnerabilities etc...

Fast ≠ Safe

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AI Governance Policies…in progress!

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South African AI Policy Incident

In 2026, South Africa withdrew its draft national AI policy after fictitious AI-generated citations were discovered.



NEWS SOUTH AFRICA

Minister Solly Malatsi's withdrawal of draft national AI policy for its fabricated sources 'embarrassing and damning'

Theolin Tembo | Published 2 weeks ago

4min

South African AI Policy Incident

Key lessons:

- Which level of AI was used?
- AI output is not verified truth
- Human review is mandatory
- Governance applies even to policymakers
- AI can accelerate mistakes at scale

"If AI can hallucinate inside AI policy itself, what happens when AI touches infrastructure?"

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AI Governance

At the CHPC, AI governance is approached in layers:

1. Organization-wide AI policy
2. Infrastructure-specific AI policy
3. Technical enforcement controls

"Policy alone is not enough. Controls must enforce policy"

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CSIR AI Policy Foundation

Broad AI governance principles include:

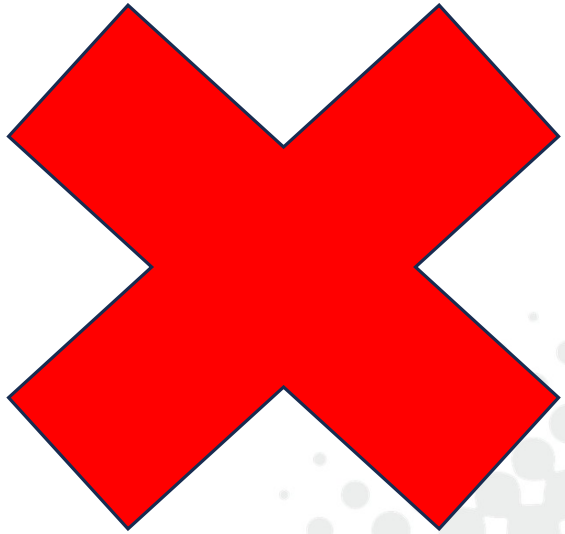
- Responsible and ethical AI use
- Human accountability remains mandatory
- Protection of sensitive data
- Legal compliance
- Transparency in AI-assisted work

“This provides the high-level governance framework.”

Why Generic AI Policy Is Not Enough

Infrastructure teams need more specific rules:

- Can AI access SSH credentials?
- Can AI trigger deployments?
- Can AI access production databases?
- Can AI execute privileged commands?



Infrastructure needs explicit guardrails.

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Data Sovereignty Matters

AI governance is not only about access control.

It is also about:

- Where data lives
- Where computation happens
- Which legal jurisdiction applies
- Who controls the infrastructure

“Key question…Where is your AI processing your data?”

Convenient ≠ Safe

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CHPC AI Policy

Infrastructure-focused AI policy includes direct operational rules:

- No AI agent/tool may store or access private SSH keys
- No AI agent/tool may receive root or sudo credentials
- No autonomous production changes without approval
- No AI agent/tool may access sensitive user data by default
- All AI actions must be logged and auditable

These are easier to enforce for staff but more challenging for users

CHPC AI Policy

These controls can be directly enforced for infrastructure staff through technical and administrative controls.

For end users, complete enforcement is not possible!

Therefore, the infrastructure rules applies the following to everyone:

- Least-privilege access
- Network Isolation
- Monitoring and auditing
- Promoting secure AI usage through awareness

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Support Local Infrastructure!

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Local Infrastructure

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With local HPC or cloud infrastructure, we can support:

- Data remaining in South Africa
- Better control over storage and access
- Compliance with local governance requirements
- Reduced exposure to foreign legal frameworks…geopolitics

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"We provide secure infrastructure you fully manage"

Practical Application: Try Local Infrastructure!

A practical alternative is to run VMs, databases, or AI workloads **on local South African infrastructure.**



Through CHPC infrastructure you have multiple options:

1. HPC Cluster Lengau – CPU and GPU Compute
2. Sebowa Cloud – Openstack Self Service Cloud Portal
3. **VuthaStack – Spin up VM's and pre-built stacks with in minutes**

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VuthaStack: VMs in minutes

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Quickly evaluate services before fully comprehensive Sebowa Openstack environment

Examples:

- VM Only: Ubuntu, Rocky Linux, CentOS VM
- GPU-enabled (NVIDIA A100) VM, and development sandbox
- Pre-built stacks: Chatbot, JupyterLab, Kubernetes
- Open WebUI (Qwen3.6, gemma4, qwen3)



This gives users hands-on access to South African infrastructure in a controlled environment.

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VuthaStack
A CHPC cloud initiative

Why VuthaStack | Services | Pricing | Request access

Powered by Sebowa infrastructure

Cloud access. In minutes.

VuthaStack is a proposed self-service cloud experience from the CSIR Centre for High Performance Computing. Start with a secure, ready-to-use virtual machine without first having to configure every part of the cloud environment yourself.

Explore pricing → | View services

✓ Hosted in South Africa ✓ Local technical support ✓ Built on Sebowa

Platform preview

Launch your first virtual machine

A simple starting point for most clients.

OPERATING SYSTEM

Ubuntu 24.04 LTS

INSTANCE SIZE

Starter VM

1 vCPU	2 GB	100 GB
Compute	Memory	Storage

Launch VM

Closing

Responsible AI infrastructure requires three pillars:

- Governance: what AI is allowed to do
- Security: what AI is allowed to access
- Sovereignty: where AI is allowed to run

Through supporting local infrastructure, AI governance becomes practical - not just policy.

In infrastructure, “trust” must be earned — never assumed.

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Thank You!

"Decentralize and support local!"

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